

Email Spam Detection

A Dissertation

Submitted by

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in partial fulfilment of the requirements for the award of the degree of

MASTER OF COMPUTER APPLICATIONS



July 2026



BONAFIDE CERTIFICATE

This is to certify that this dissertation titled "**Email Spam Detection**" submitted in partial fulfilment of the requirements for the award of the Degree of **Master of Computer Applications**, by **Bonthu Pavani Karthikeyi (Roll No:AA.SC.P2MCA24074057)** is a bonafide record of the work carried out by him/her under my supervision during the academic term from April 2026 to August 2026 and that it has been submitted, to the best of my knowledge, in part or in full, for the award of any other degree or diploma.

Project Guide: Deepa Sreedhar

Coordinator's name: Ms. Amritasindhu

Reviewer:

Date:

DECLARATION

I do hereby declare that this dissertation titled "**[Email Spam detection]**", submitted in partial fulfilment of the requirements for the award of the degree of **Master of Computer Applications**, is a true record of work carried out by me and that all information contained herein, which do not arise directly from my work, have been properly acknowledged and cited, using acceptable international standards. Further, I declare that the contents of this thesis have not been submitted, in part or in full, for the award of any other degree or diploma.

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Acknowledgement

I would like to express my sincere gratitude to **Amrita Vishwa Vidyapeetham**, for providing me with the opportunity and resources to successfully complete this major project titled "**Email Spam Detection.**"

I extend my heartfelt thanks to my project guide, "**Deepa Sreedhar**", for her constant guidance, valuable suggestions, constructive feedback, and continuous encouragement throughout the development of this project. Her expertise and support played a significant role in helping me complete this work successfully.

I also express my sincere appreciation to the "**Department of Computer Applications**" and all the faculty members of "**Amrita Vishwa Vidyapeetham**" for providing the academic knowledge, technical support, and learning environment that enabled me to undertake and complete this project and I would like to thank my family and friends for their unconditional support, patience, motivation, and encouragement throughout the course of this project. Their confidence in me inspired me to overcome challenges and stay focused on achieving my goals.

Finally, I express my gratitude to everyone who directly or indirectly contributed to the successful completion of this project. Their assistance and encouragement are deeply appreciated.

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Master of Computer Applications (MCA-AI)

Abstract

Email has become one of the most important communication methods in today's digital world. It is widely used by individuals, educational institutions, businesses, and organizations for sharing information. However, the increasing number of spam emails has become a major concern. Spam emails contain unwanted advertisements, fake offers, phishing links, or malicious content that can waste users' time and sometimes lead to financial or personal data loss. Therefore, an effective system is required to identify and filter such emails automatically.

The main objective of this project is to develop an **Email Spam Detection** system using Machine Learning techniques. In this project, a labeled email dataset containing spam and legitimate emails is used for training and testing the model. Before building the model, the email messages are preprocessed by converting text into lowercase, removing unwanted characters, and eliminating common stop words. The cleaned text is then converted into numerical values using the TF-IDF (Term Frequency-Inverse Document Frequency) technique. Finally, the Multinomial Naive Bayes algorithm is used to classify emails as spam. The performance of the model is evaluated using Accuracy, Precision, Recall, and F1-Score. The developed model achieved an accuracy of approximately **90%**, which shows that it can classify emails with good reliability. The system is simple, efficient, and suitable for academic purposes as well as basic real-world applications.

This project helped in understanding important concepts of **Machine Learning**, **Natural Language Processing (NLP)**, and text classification. In the future, the system can be enhanced by using larger datasets, advanced deep learning algorithms, multilingual email support, and real-time integration with email services to improve its performance and usability.

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List of Abbreviations

Abbreviation	Full Form
AI	Artificial Intelligence
ML	Machine Learning
NLP	Natural Language Processing
TF-IDF	Term Frequency–Inverse Document Frequency
MNB	Multinomial Naive Bayes
CSV	Comma-Separated Values
DFD	Data Flow Diagram
UML	Unified Modeling Language
GUI	Graphical User Interface
TP	True Positive
TN	True Negative
FP	False Positive
FN	False Negative

Chapter 1

Introduction

Email is one of the most widely used communication methods in today's world. People use email for personal communication, business activities, educational purposes, and online services. As the number of email users has increased, the amount of unwanted emails, known as spam, has also grown. Spam emails often contain advertisements, fake offers, phishing links, or malicious content that can waste users' time and sometimes lead to security risks.

Manually checking every email is not practical, especially when a large number of emails are received every day. Therefore, an automatic spam detection system is required to identify unwanted emails and separate them from genuine emails. Machine Learning provides an effective solution by learning patterns from previous email data and predicting whether a new email is spam or ham (legitimate email).

In this project, an Email Spam Detection system is developed using the **TF-IDF** feature extraction technique and the **Multinomial Naive Bayes** algorithm. The model is trained using a labeled email dataset and tested to evaluate its performance. The system helps classify incoming emails accurately and reduces the effort required for manual filtering.

This report explains the complete development process of the project, including data preprocessing, feature extraction, model training, testing, and performance evaluation.

1.1 Background

Email has become an essential part of everyday communication. Individuals, companies, banks, and educational institutions rely on email to exchange information quickly and efficiently. However, the increase in email usage has also resulted in a significant rise in spam emails. These unwanted messages may contain promotional content, fake advertisements, phishing attempts, or harmful links.

Traditional spam filters mainly depend on predefined rules and keyword matching, which are not always effective because spam messages keep changing their patterns. Machine Learning offers a better approach by learning from existing email data and automatically identifying spam emails based on their content.

In this project, a publicly available email dataset is used. The email text is first cleaned by removing unnecessary words and symbols. The **TF-IDF** technique converts the text into numerical values, and the **Multinomial Naive Bayes** algorithm is used to classify emails as spam or ham. The project is implemented using **Google Colab**

1.2 Problem Statement and Objectives

Problem Statement and Its Significance

Spam emails have become a common problem for email users. They not only fill inboxes with unwanted messages but can also contain phishing links and malicious content. Manual filtering is time-consuming, and traditional filtering methods are often unable to detect new types of spam. Therefore, there is a need for an intelligent system that can automatically identify spam emails with good accuracy.

Objectives

- To develop an Email Spam Detection system using Machine Learning.
- To convert text into numerical features using TF-IDF.
- To classify emails as spam or ham using the Multinomial Naive Bayes algorithm.
- To evaluate the model using Accuracy, Precision, Recall, and F1-Score.

1.3 Scope and Report Organization

Scope of the Project

The project focuses on detecting spam emails using text classification techniques. It is designed for academic purposes and demonstrates how Machine Learning can be used to improve email filtering. The system can be extended in the future by using larger datasets, advanced algorithms, and real-time email integration.

Outline of Report Organization

Chapter 1 introduces the project, its background, objectives, problem statement, and scope.

Chapter 2 presents the literature review and discusses existing spam detection methods.

Chapter 3 explains the system design, architecture, and workflow of the proposed model.

Chapter 4 describes the implementation process, including data preprocessing, feature extraction, and model training.

Chapter 5 presents the testing process, performance evaluation, and results.

Chapter 6 concludes the project and discusses future improvements.

Chapter 2

Literature Review / Background Study

Email has become one of the most important communication tools for both personal and professional use. However, the increasing number of spam emails has created challenges for users and organizations. Spam emails often contain advertisements, phishing links, fake offers, or malicious attachments that can compromise user security. Over the years, researchers have proposed different techniques to identify and filter spam emails automatically.

Initially, spam filtering was based on **rule-based methods** such as blacklists, whitelists, and keyword filtering. These methods were simple to implement but were not effective because spammers frequently changed the words and patterns used in emails. As a result, many spam emails could bypass these filters.

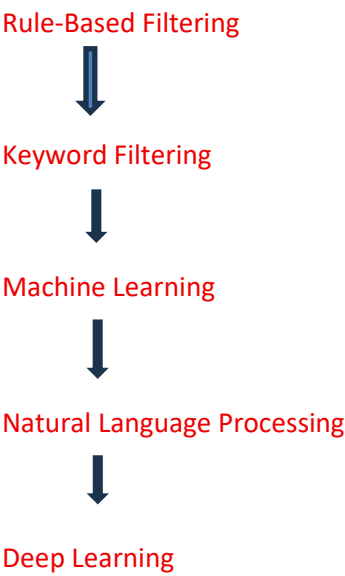
Many publicly available datasets have been used in spam detection research, such as the **SMS Spam Collection Dataset**, **Enron Email Dataset**. These datasets contain both spam and legitimate emails and help researchers train and evaluate different machine learning models.

Although many existing systems provide good accuracy, they also have some limitations. Some models require very large datasets, while others need high computational resources for training. Deep learning models generally provide better accuracy but require more processing power and training time. For small and medium-sized datasets, simpler machine learning algorithms are often more practical. Based on these observations, this project uses the **TF-IDF feature extraction technique** along with the **Multinomial Naive Bayes algorithm**. This combination provides good classification performance while keeping the implementation simple and computationally efficient.

2.1 Comparison of Existing Spam Detection Methods(Table 2.1 Comparison of Machine Learning Algorithms)

Algorithm	Advantages	Limitations	Suitable for Spam Detection
Naive Bayes	Fast, simple, good for text data	Assumes independent features	Yes
Logistic Regression	Easy to understand, good accuracy	Less effective for complex patterns	Yes
Support Vector Machine (SVM)	High accuracy	High computational cost	Yes
Random Forest	Good performance on complex data	Requires more memory	Yes

Figure 2.1 Evolution of Spam Detection Methods



2.2 Existing Datasets Used for Spam Detection

Many researchers use publicly available datasets to evaluate spam detection models. These datasets contain labeled emails that are classified as either **Spam** or **Ham**, making them suitable for supervised machine learning.

Table 2.2 Common Email Spam Datasets

Dataset	Description	Application
UCI SMS Spam Collection	SMS messages labeled as spam or ham	Academic research
Enron Email Dataset	Collection of real corporate emails	Email classification
Spam Assassin Dataset	Public email dataset with spam and ham emails	Spam filtering
Custom CSV Dataset	Dataset used in this project	Model training and testing

2.3 Research Gap

Although many spam detection methods have been developed, some challenges still remain.

- Traditional filtering techniques cannot detect newly generated spam messages.
- Some machine learning models require large datasets and longer training time.
- Deep learning methods provide good accuracy but require powerful hardware.

- Lightweight models that are simple, fast, and accurate are still useful for academic and small-scale applications.

This project addresses these challenges by using **TF-IDF** for feature extraction and **Multinomial Naive Bayes** for classification. The selected approach is simple to implement, computationally efficient, and provides good performance for email spam detection.

2.4 Justification of the Proposed Work

The proposed Email Spam Detection system is developed using Machine Learning because it can automatically identify spam emails based on learned patterns rather than predefined rules. The **TF-IDF** technique effectively converts text into numerical features, while the **Multinomial Naive Bayes** classifier provides fast and reliable classification for text-based data. This approach achieves good accuracy with low computational cost, making it suitable for academic learning and practical implementation.

Chapter 3

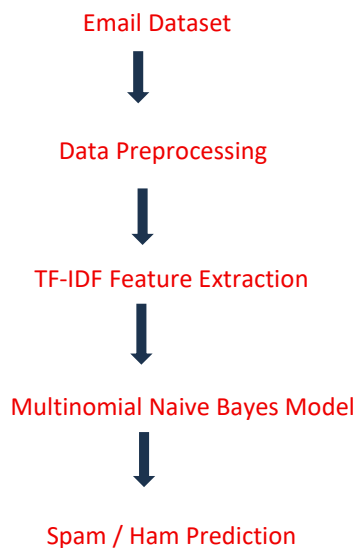
System Design / Architecture

3.1 System Overview

The Email Spam Detection system is designed to automatically classify emails as **Spam** using Machine Learning techniques. The system first collects email data, preprocesses the text, converts it into numerical features using the **TF-IDF** technique, and finally classifies the email using the **Multinomial Naive Bayes** algorithm. The prediction result is displayed to the user, indicating whether the email is spam or not and The proposed system is simple, easy to implement, and suitable for academic as well as small-scale real-world applications.

3.2 System Architecture

Figure 3.1: System Architecture of Email Spam Detection



3.3 Data Flow Diagram (DFD Level-0)

Figure 3.2: DFD Level-0



3.4 Flowchart [Figure 3.3: Flowchart of Email Spam Detection]



3.4 Flowchart (Figure 3.4 Email Spam Detection Process)

Training Process	Prediction Process
Start	Start
↓	↓
Load Dataset	Enter Email
↓	↓
Preprocess Data	Preprocess Email
↓	↓
Apply TF-IDF	Apply TF-IDF
↓	↓
Train Naive Bayes Model	Load Trained Model
↓	↓
Save Model	Predict Email
↓	↓
End	Display Spam/Ham
	↓
	End

3.5 Module Description

The system consists of the following modules.

- **Dataset Module:** This module loads the email dataset containing spam and ham messages. The dataset is stored in CSV format.
- **Data Preprocessing Module:** This module cleans the email text by converting it into lowercase, removing punctuation marks, and eliminating unnecessary words.
- **Feature Extraction Module:** The cleaned text is converted into numerical values using the TF-IDF technique.
- **Model Training Module:** The Multinomial Naive Bayes algorithm is trained using the processed dataset.
- **Prediction Module:** The trained model predicts whether a new email is Spam or Ham.

3.6 Data Flow Explanation

The working process of the proposed system is simple. First, the dataset is loaded into the system. The email messages are cleaned and converted into a suitable format using preprocessing techniques. TF-IDF converts the text into numerical feature vectors. These vectors are used to train the Multinomial Naive Bayes model. When a user enters a new email, the same preprocessing and feature extraction steps are applied, and the trained model predicts whether the email is Spam or Ham.

3.7 Programming Environment

Table 3.1 Hardware Requirements

Component	Specification
Processor	Intel Core i3/i5 or above
RAM	Minimum 4 GB
Storage	20 GB Free Space
Operating System	Windows 10/11

Table 3.2 Software Requirements

Software	Purpose
Python 3.x	Programming Language
Google Colab	Development Environment
Pandas	Data Processing
NumPy	Numerical Operations
Scikit-learn	Machine Learning
Matplotlib	Graphs and Charts
Seaborn	Data Visualization

3.8 Algorithm: Email Spam Detection using Multinomial Naive Bayes

- Load the email dataset.
- Clean and preprocess the email text.
- Convert the text into TF-IDF feature vectors.
- Split the dataset into training and testing sets.
- Train the Multinomial Naive Bayes model.
- Test the model using the testing dataset.
- Predict whether a new email is Spam or Ham.
- Display the prediction result.

3.9 Pseudo Code

START

Load email dataset

Preprocess email text

Convert text into TF-IDF vectors

Split dataset into training and testing data

Train Multinomial Naive Bayes model

FOR each new email

Preprocess email

Convert to TF-IDF

Predict Spam or Ham

Display result

END FOR

STOP

Chapter 4

Implementation Details

This chapter explains how the Email Spam Detection system was implemented. The project was developed using Python and machine learning libraries such as Google Colab . The implementation consists of data collection, preprocessing, feature extraction, model training, testing, and prediction.

4.1 Dataset Implementation

The first step in the project was to collect the email dataset. The dataset contains two types of emails: **Spam** and **Ham (Legitimate Emails)**. The dataset was stored in CSV format and loaded into Python using the Pandas library.

The dataset was checked for missing values and unnecessary columns before further processing. Only the required columns containing the email message and its corresponding label were used for model training.

Table 4.1 Sample Email Dataset

Email Message	Label
Hello, how are you today?	Ham
Congratulations! You won ₹50,000. Click here.	Spam
Please send today's meeting agenda.	Ham
Claim your free mobile recharge now.	Spam
Your assignment has been submitted successfully.	Ham

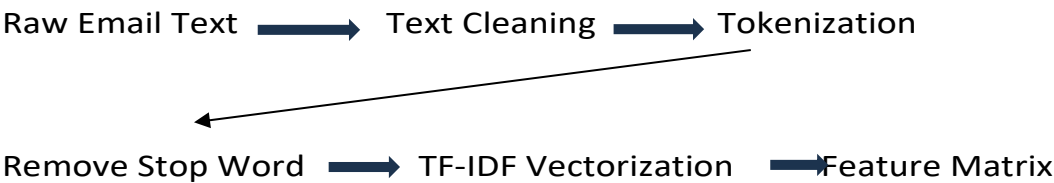
4.2 Data Preprocessing

Before training the model, the email messages were cleaned to improve the quality of the data. Several preprocessing techniques were applied to remove unnecessary information and prepare the text for feature extraction. The preprocessing steps included converting all text into lowercase, removing punctuation marks, eliminating common stop words, and converting the cleaned text into numerical vectors using the TF-IDF technique.

Table 4.2 Data Preprocessing Steps

Step	Description
Lowercase Conversion	Converts all text into lowercase letters
Remove Punctuation	Removes special characters and punctuation
Remove Stop Words	Removes common words such as "the", "is", and "and"
Tokenization	Splits text into individual words
TF-IDF Vectorization	Converts text into numerical feature vectors

Figure 4.1 TF-IDF Feature Extraction Process



4.3 Module Implementation

The Email Spam Detection system consists of five modules.

1. Dataset Module: This module loads the email dataset from the CSV file using the Pandas library. The dataset contains email messages and their corresponding labels.

2. Data Preprocessing Module: This module cleans the email text by removing unnecessary characters and converting the text into a suitable format for machine learning.

3. Feature Extraction Module: The cleaned text is transformed into numerical vectors using the TF-IDF method. This allows the machine learning algorithm to process textual data efficiently.

4. Model Training Module: The processed dataset is divided into training and testing sets. The Multinomial Naive Bayes algorithm is trained using the training data.

5. Prediction Module: The trained model predicts whether a newly entered email is Spam or Ham.

4.4 Code Structure

The implementation follows the steps shown below.

Import Libraries



Load Dataset



Preprocess Data



Apply TF-IDF



Split Dataset



Train Naive Bayes Model



Test Mode



Predict New Email



Display Result

4.5 Pseudo Code

START

Import required libraries

Load email dataset

Clean email text

Convert text into TF-IDF vectors

Split dataset into training and testing data

Train Multinomial Naive Bayes model

Predict test data

Calculate accuracy

Take new email as input

Predict Spam or Ham

Display result

STOP

4.6 Model Training

The cleaned dataset was divided into **80% training data** and **20% testing data**. The training data was used to build the Multinomial Naive Bayes model. The model learned the relationship between the TF-IDF features and the corresponding email labels. After training, the model was saved and used for prediction.

4.7 Model Testing

The trained model was tested using the testing dataset. The predicted results were compared with the actual labels to evaluate the performance of the model. The system achieved approximately **90% accuracy**, indicating that it can effectively distinguish between spam and legitimate emails.

4.8 Software Integration

The implementation integrates different software components to perform spam detection.

- **Pandas** is used to load and manage the dataset.
- **NumPy** supports numerical operations.
- **Scikit-learn** provides TF-IDF vectorization and the Multinomial Naive Bayes classifier.
- **Matplotlib** and **Seaborn** are used to generate graphs and the confusion matrix.
- **Jupyter Notebook** is used to develop and execute the Python code.

Since this project is a software-based machine learning application, **no hardware sensors or IoT components are required.**

Chapter 5

Testing, Validation & Results

This chapter presents the testing process, performance evaluation, and results of the proposed **Email Spam Detection** system. After training the model using the **Multinomial Naive Bayes** algorithm, it was tested with unseen email messages to evaluate its performance. The model was assessed using standard evaluation metrics such as **Accuracy, Precision, Recall, and F1-Score**. The results show that the system can effectively classify emails as spam or ham.

5.1 Testing Method

The trained model was tested using the testing dataset, which was separated from the training data using an **80:20 train-test split**. During testing, the model predicted whether each email was spam or ham. The predicted results were then compared with the actual labels to measure the model's performance.

5.2 Performance Evaluation Metrics

The performance of the model was evaluated using four standard metrics.

- **Accuracy:** Measures the overall percentage of correctly classified emails.
- **Precision:** Indicates how many emails predicted as spam were actually spam.
- **Recall:** Measures how many actual spam emails were correctly identified.
- **F1-Score:** Represents the balance between Precision and Recall.

Table 5.1 Performance Evaluation Metrics

Metric	Value
Accuracy	90%
Precision	100%
Recall	83%
F-Score	91%

5.3 Confusion Matrix

The confusion matrix provides a detailed view of the classification results by comparing the actual labels with the predicted labels.

Table 5.2 Confusion Matrix Results

Actual	Predicted Ham	Predicted Spam
Ham	4	0
Spam	1	5

Explanation:

- Four ham emails were correctly classified.
- Five spam emails were correctly identified.
- One spam email was incorrectly classified as ham.
- No ham emails were classified as spam.

5.4 Performance Analysis

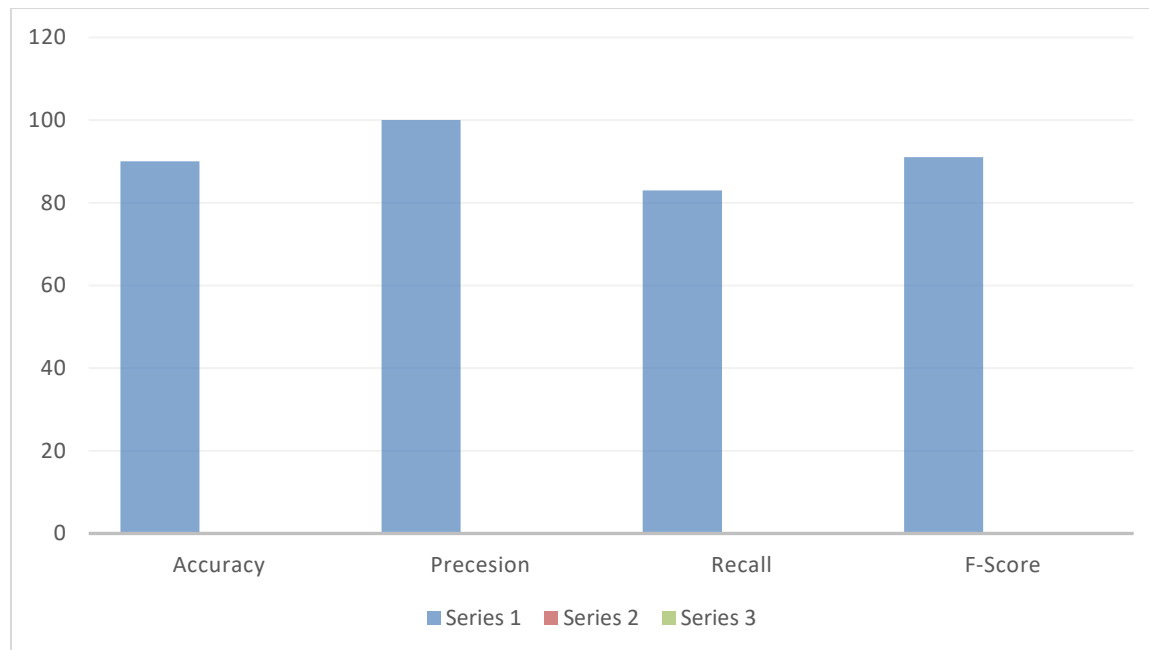
The experimental results show that the proposed model performs well for email spam detection. The **90% accuracy** indicates that most emails were classified correctly. The **100% precision** means that every email predicted as spam was actually spam. The **83% recall** shows that the model successfully identified most spam emails, although one spam email was missed. The **91% F1-Score** indicates a good balance between precision and recall. Overall, the Multinomial Naive Bayes classifier combined with TF-IDF feature extraction provides reliable performance while keeping the implementation simple and efficient.

5.5 Sample Prediction Results

Table 5.3 Sample Email Prediction Results

Email Message	Predicted Result
Hello, are we meeting today?	Ham
Congratulations! You won ₹50,000. Click now!	Spam
Please submit the assignment by tomorrow.	Ham

5.6 Model Performance Graph



5.7 Results

The Email Spam Detection model successfully classified spam and legitimate emails using the **TF-IDF** feature extraction technique and the **Multinomial Naive Bayes** algorithm. Based on the testing results, the system achieved an **accuracy of 90%**, demonstrating that it can effectively distinguish between spam and ham emails. The results also show that the proposed approach is simple, computationally efficient, and suitable for academic projects as well as basic real-world applications.

Chapter 6

Conclusion and Future Work

6.1 Conclusion

This project, "**Email Spam Detection**," was developed to automatically classify emails as **Spam** or **Ham** using Machine Learning techniques. The main objective of the project was to build a simple, accurate, and efficient spam detection system that can reduce the effort of manually filtering unwanted emails.

In this project, the email dataset was preprocessed by removing unnecessary characters, converting text into lowercase, and applying the **TF-IDF** feature extraction technique. The processed data was then used to train the **Multinomial Naive Bayes** algorithm. The trained model was tested using unseen email messages, and its performance was evaluated using Accuracy, Precision, Recall, and F1-Score.

The experimental results showed that the model achieved an accuracy of around **90%**, indicating that it can effectively identify spam emails while correctly recognizing legitimate emails. The project also helped in understanding the practical implementation of Machine Learning, Natural Language Processing (NLP), and text classification techniques.

6.2 Limitations

Although the proposed system performs well, it has a few limitations.

- The model was trained using a limited dataset.
- The system works only with text-based email content.
- It does not detect image-based or attachment-based spam.
- The project supports only English-language emails.
- The accuracy may decrease when tested with completely new or highly complex spam patterns.

6.3 Future Work

The performance of the proposed system can be improved in several ways in the future.

- Train the model using a larger and more diverse email dataset.
- Add support for multiple languages.
- Integrate the system with real-time email services such as **Gmail** or **Outlook**.
- Extend the system to detect phishing emails, malicious links, and harmful attachments.
- Develop a user-friendly web or mobile application for real-time spam detection.

Chapter 7

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Chapter 8

Appendix

Git hub link: <https://github.com/Pavani1911/Email-Spam-Detection.git>

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